AI-Driven Drug Discovery

Molecular Property Analysis, Machine Learning Screening, and Graph Neural Networks

Executive Summary

This report presents an end-to-end computational drug discovery workflow combining molecular data preparation, chemical descriptor analysis, classical machine learning, and graph neural networks (GNNs). The workflow is designed to validate molecular structures, calculate interpretable physicochemical features, assess drug-likeness, train predictive models, rank candidate compounds, and export screening results for downstream analysis.

1. Project Objectives

- Clean and validate molecular structures represented as SMILES.
- Generate molecular descriptors and Morgan fingerprints for computational analysis.
- Visualize molecular structures and evaluate Lipinski Rule of Five properties.
- Train predictive models for molecular activity/property classification.
- Evaluate models using accuracy, ROC-AUC, confusion matrices, ROC curves, and macro F1 where appropriate.
- Use molecular graphs with a PyTorch Geometric GNN to learn directly from atom and bond connectivity.
- Rank compounds for batch screening and export predictions/models for later use.

2. Data Preparation and Molecular Representation

The workflow begins with molecular records represented by SMILES strings. RDKit is used to parse and validate the structures, identify invalid records, and generate molecular representations. The project used both a synthetic 600-record dataset during development and a larger 10,000-record, 12-column dataset for the broader workflow.

For each valid molecule, the workflow can generate a 2D molecular depiction and a collection of descriptors. These provide both interpretable chemical information and numerical inputs suitable for machine-learning pipelines.

3. Molecular Descriptors

Descriptor	Purpose
Molecular Weight	Measures molecular size and contributes to drug-likeness assessment.
LogP	Estimates lipophilicity and helps characterize permeability and solubility tendencies.
TPSA	Captures polar surface area and provides information relevant to permeability.
H-bond Donors (HBD)	Counts hydrogen-bond donating groups.
H-bond Acceptors (HBA)	Counts hydrogen-bond accepting atoms.
Rotatable Bonds	Provides an indication of molecular flexibility.
Aromatic Rings	Measures aromatic structural content.

4. Lipinski Rule of Five and Drug-Likeness

The workflow evaluates the common Lipinski Rule of Five criteria using molecular weight, LogP, hydrogen-bond donors, and hydrogen-bond acceptors. These rules provide a practical first-pass assessment of oral drug-likeness rather than a guarantee of biological activity, efficacy, safety, or clinical success.

5. Fingerprints and Classical Machine Learning

Morgan fingerprints encode local molecular environments as fixed-length binary or count-based representations. They are particularly useful for similarity-aware cheminformatics and machine-learning models because they capture substructural patterns that may correlate with molecular activity.

A Random Forest classifier forms the main classical machine-learning component. An optional XGBoost model can provide an additional tree-based comparison. Model evaluation includes accuracy and ROC-AUC, together with confusion matrices and ROC curves where applicable. Feature-importance analysis helps identify descriptors associated with predictions.

6. Molecular Graph Neural Network

The dedicated GNN module represents each molecule as a graph: atoms are nodes and chemical bonds are edges. Atom and bond features are supplied to a molecular graph convolutional network implemented with PyTorch Geometric. This representation allows the model to learn from molecular connectivity instead of relying only on manually selected global descriptors.

The training workflow includes training/validation monitoring and early stopping to reduce overfitting. Evaluation includes test accuracy, macro F1, and a confusion matrix. Training and validation curves provide an additional diagnostic view of model learning and generalization.

7. Screening and Prediction Workflow

1. Validate and clean input molecules.
2. Calculate descriptors and fingerprints.
3. Apply drug-likeness filters and exploratory visualization.
4. Train and evaluate classical ML models.
5. Train/evaluate the molecular GNN.
6. Generate predictions for new or batch molecules.
7. Rank candidate compounds according to predicted activity/probability.
8. Export predictions to CSV and save trained model artifacts for reuse.

8. Key Findings and Interpretation

The combined workflow demonstrates the value of using multiple molecular representations. Descriptor-based models are relatively interpretable and computationally efficient, while Morgan fingerprints capture substructural patterns that may not be represented by a small descriptor set. The GNN provides a complementary approach by learning directly from graph structure and atom/bond features.

The most useful outputs for candidate prioritization are the model prediction scores, hit rankings, molecular descriptors, drug-likeness indicators, and structural representations. These outputs can be reviewed together rather than treating a

single model score as a definitive indication of biological activity.

9. Reproducibility and Export

The project is structured so that molecular validation, descriptor calculation, model training, prediction, and screening can be repeated on new datasets. CSV prediction outputs and trained-model files support downstream analysis and reuse. The GNN component likewise supports saving the trained model and prediction results.

10. Limitations and Next Steps

- Predictions depend strongly on dataset quality, class balance, labeling, and chemical diversity.
- Drug-likeness rules are heuristic filters and should not be interpreted as clinical evidence.
- Model metrics should be evaluated on carefully separated validation/test data to avoid overly optimistic estimates.
- External validation and prospective experimental testing are required before biological or therapeutic conclusions can be made.
- Future improvements may include broader molecular features, hyperparameter optimization, explainable GNN methods, uncertainty estimation, and external datasets.

Conclusion

The drug discovery pipeline integrates cheminformatics, classical machine learning, and graph-based deep learning into a single screening framework. RDKit-based validation and descriptors provide an interpretable foundation; Morgan fingerprints and Random Forest/XGBoost provide strong classical baselines; and the PyTorch Geometric GNN adds a structure-aware learning approach. Together, these components create a practical computational framework for prioritizing molecules for further study.